

Cyber Security

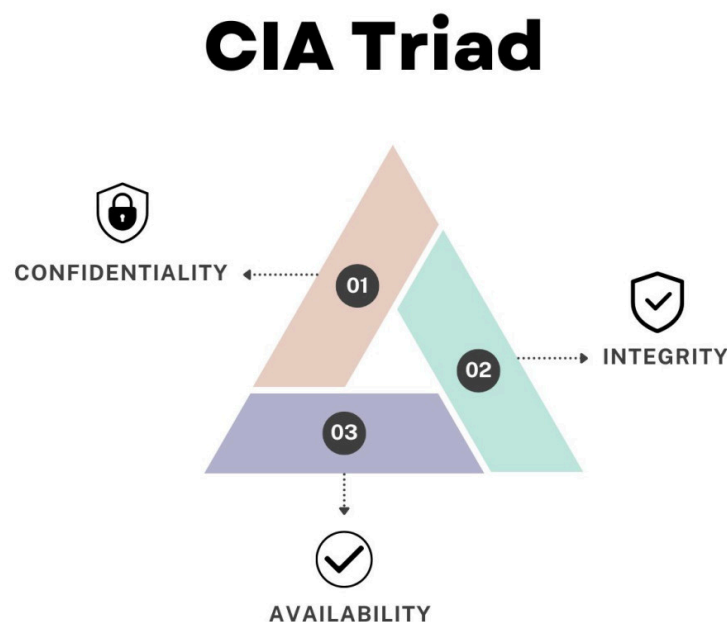
Basic Concepts of Threats, Vulnerabilities, Controls; Risk

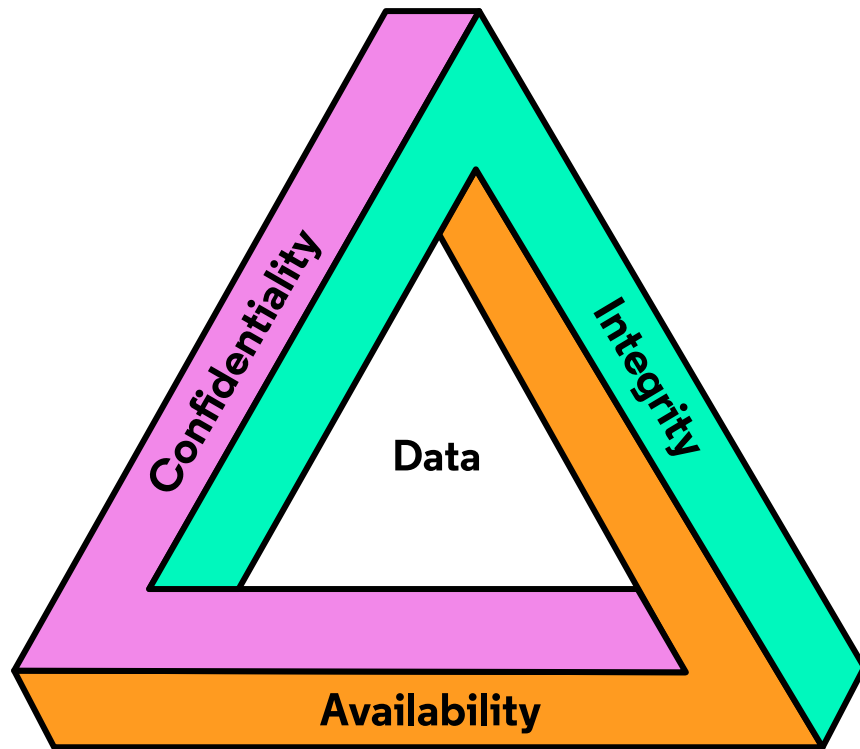
- **Threat** → Any potential danger that can exploit a system (e.g., hackers, viruses, phishing).
- **Vulnerability** → A weakness in software/hardware that can be exploited (e.g., outdated software, weak password).
- **Controls** → Measures to reduce risks (e.g., antivirus, firewalls, strong passwords).
- **Risk** → The likelihood + impact of a threat exploiting a vulnerability (Risk = Threat × Vulnerability × Impact).

Confidentiality, Integrity, Availability (CIA Triad)

The **CIA Triad** is the foundation of information security:

Here are clear diagrams explaining the **CIA Triad** (Confidentiality, Integrity, Availability):





- **Integrity** → Data remains accurate and unchanged (e.g., no tampering by hackers).
- **Availability** → Data and systems are accessible when needed (e.g., protection from DDoS attacks).

Security Policies & Mechanisms

- **Security Policies** → Rules set by organizations/individuals (e.g., "use strong passwords", "don't share login details").
- **Security Mechanisms** → Tools/techniques like encryption, authentication (passwords, biometrics), access controls.

Data Security and Protection

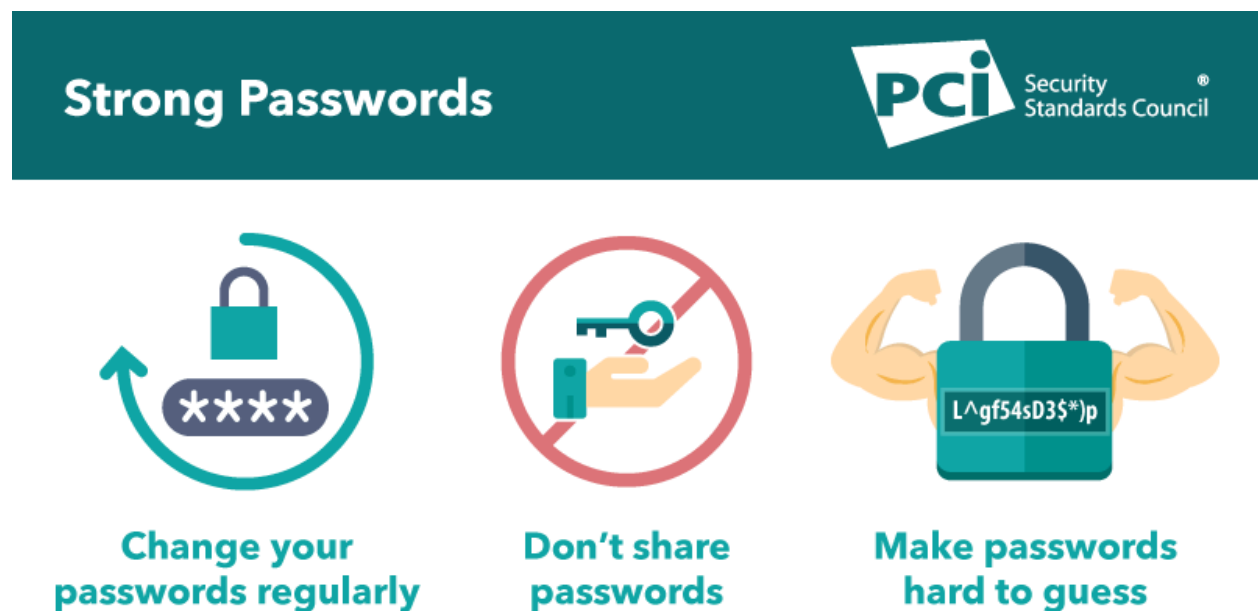
Data Security means protecting data from unauthorized access, loss, or damage.

Creating Strong Passwords

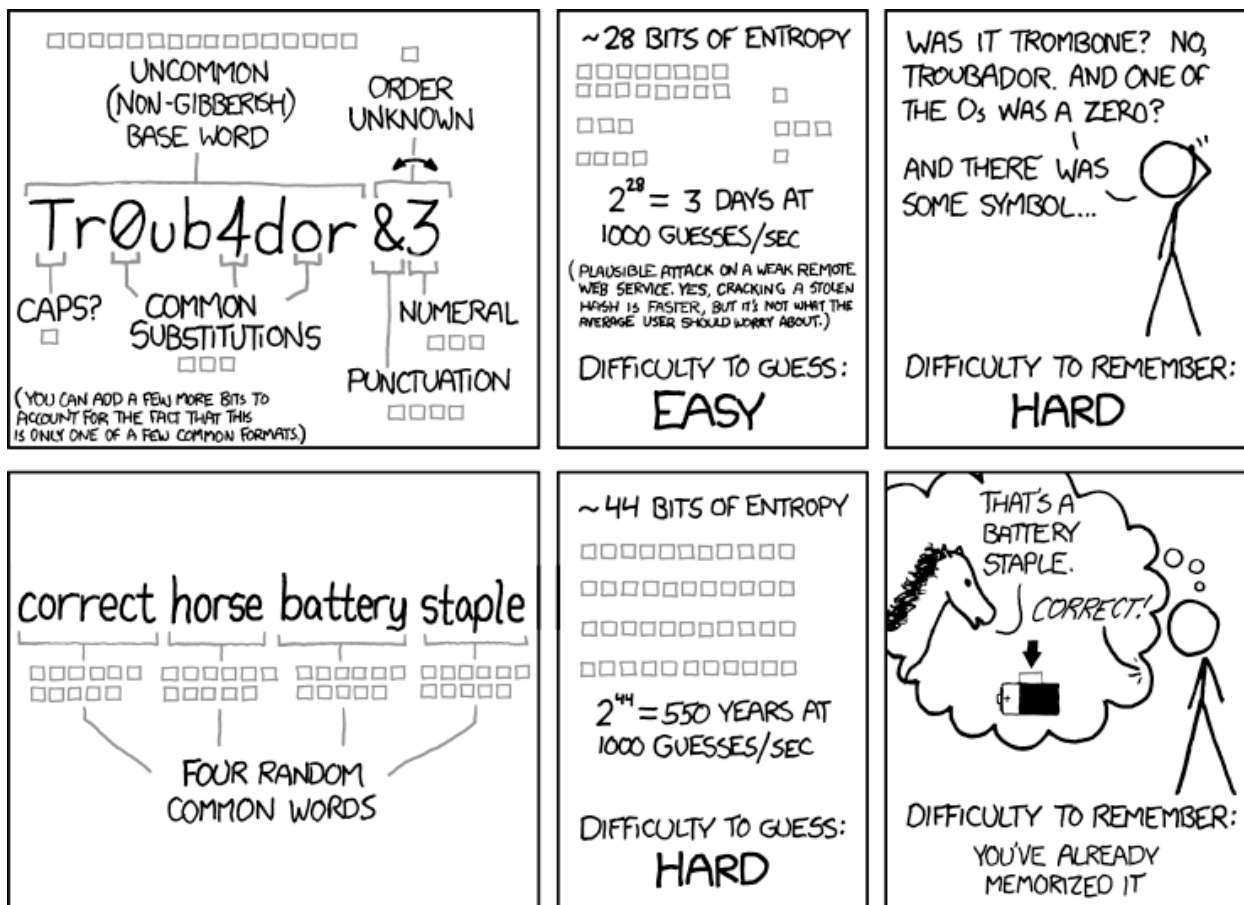
Use:

- At least 12-16 characters.
- Mix of uppercase, lowercase, numbers, symbols.
- Avoid common words (e.g., don't use "password123" or your name).

Here are helpful infographics with **strong password creation tips**:



blog.pcisecuritystandards.org



THROUGH 20 YEARS OF EFFORT, WE'VE SUCCESSFULLY TRAINED EVERYONE TO USE PASSWORDS THAT ARE HARD FOR HUMANS TO REMEMBER, BUT EASY FOR COMPUTERS TO GUESS.

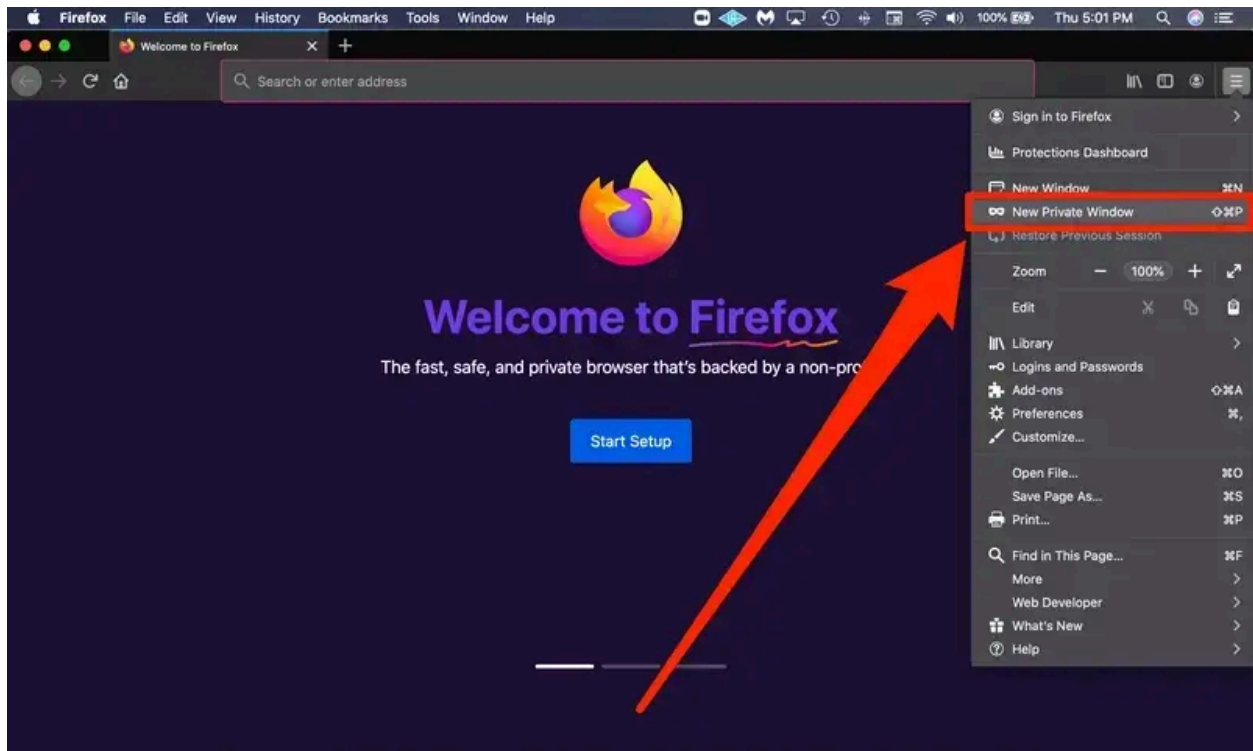
inspiredelearning.com

Tips: Use a password manager, enable two-factor authentication (2FA), change passwords regularly.

How to Stay Safe When Surfing on the Internet

- Use **In Private Browsing** (Incognito mode) → Doesn't save history, cookies, or form data (good for public computers).

Here are examples of **private browsing / incognito mode** icons in different browsers:

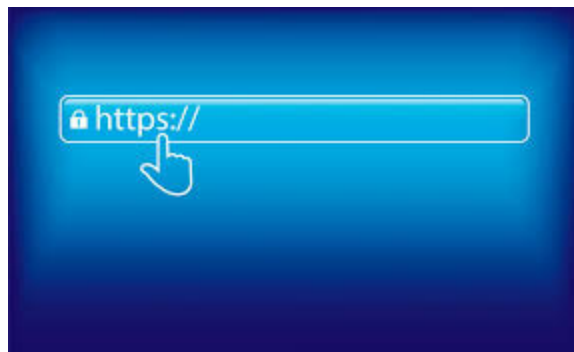


[businessinsider.com](https://www.businessinsider.com)

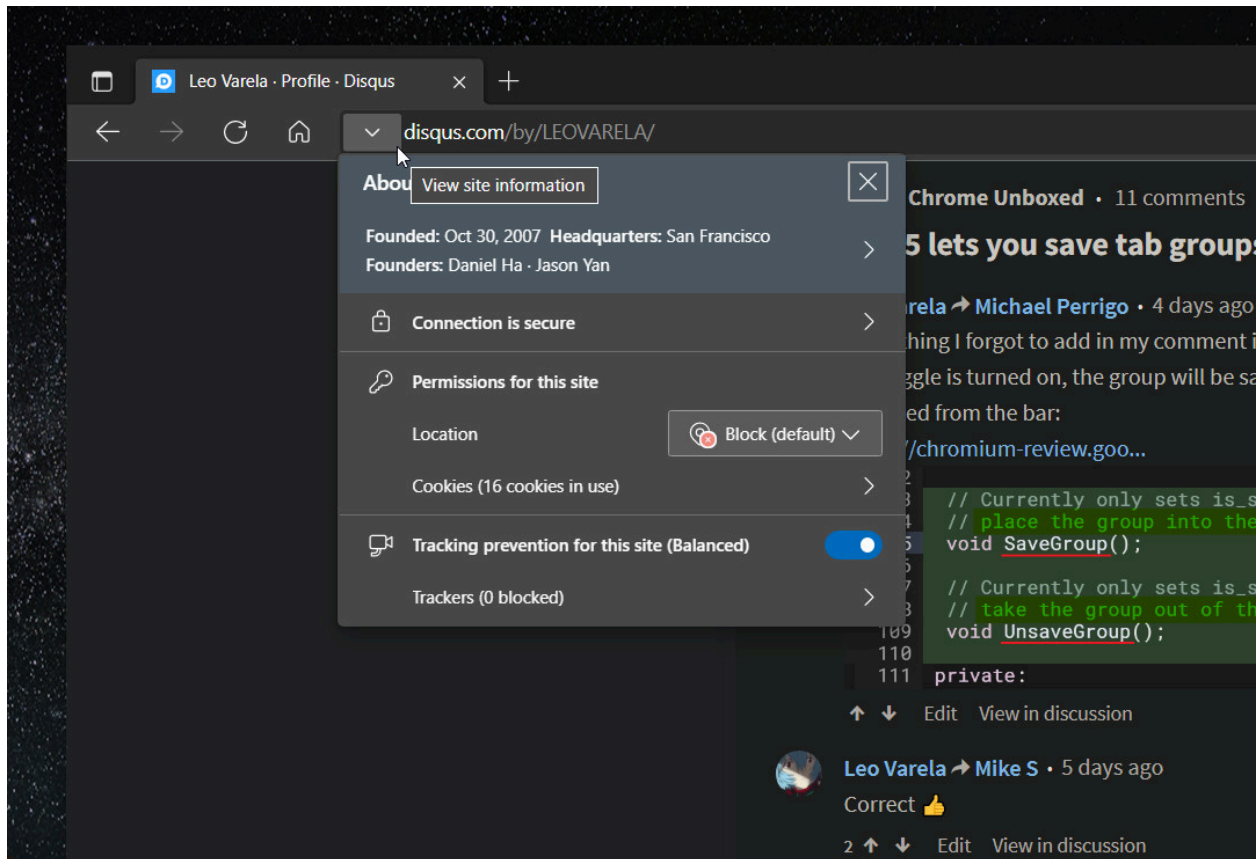
How to Use Incognito Mode: Chrome, Safari, Edge, Firefox

- **Identify Secure Websites** → Look for:
 - **https://** (not http://) in the address bar.
 - **Padlock (lock) symbol** next to the URL (means encrypted connection via SSL/TLS).

Here are real examples of the **HTTPS lock symbol** in browser address bars (Chrome, Edge, etc.):



inacom-sby.com



neowin.net

- **Clear Cookies** → Regularly clear browsing data (Settings → Privacy → Clear browsing data) to remove tracking cookies.

Security Considerations: Threats from Websites (Malware)

Malware (Malicious Software) includes harmful programs that damage or steal data.

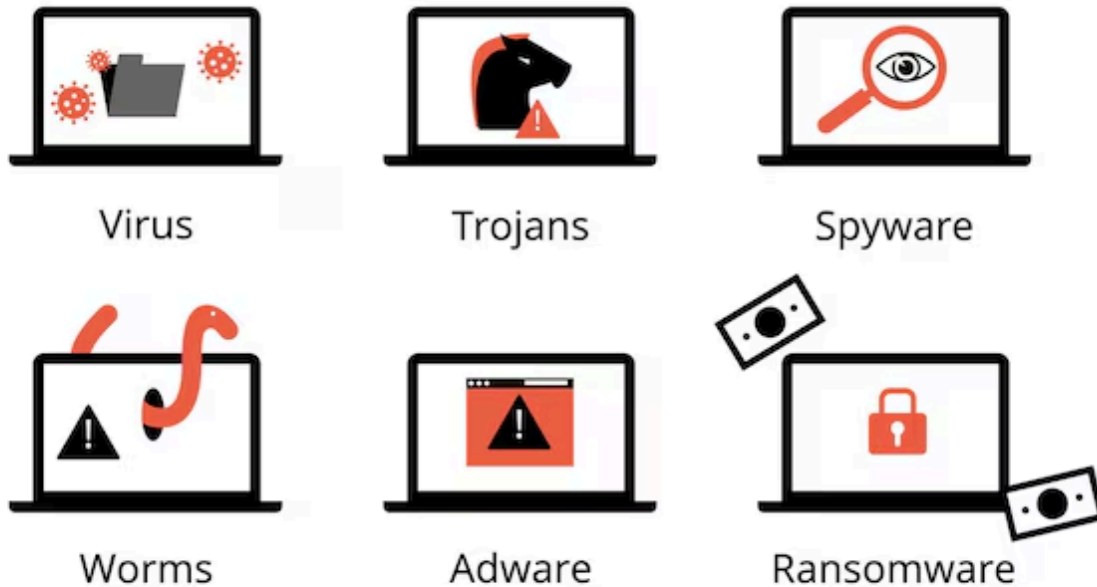
Common types:

Here are illustrations showing different **malware types** (viruses, worms, Trojans, spyware):



cheq.ai

Types of Malware



[freepik.com](https://www.freepik.com)

- **Virus** → Attaches to files, spreads when file is opened.
- **Worm** → Self-replicates across networks without user action.
- **Trojan Horse** → Pretends to be useful software but contains hidden malicious code.
- **Spyware** → Secretly monitors and steals information (e.g., keystrokes, passwords).

Protect yourself: Use updated antivirus (e.g., Windows Defender), avoid suspicious downloads/links.

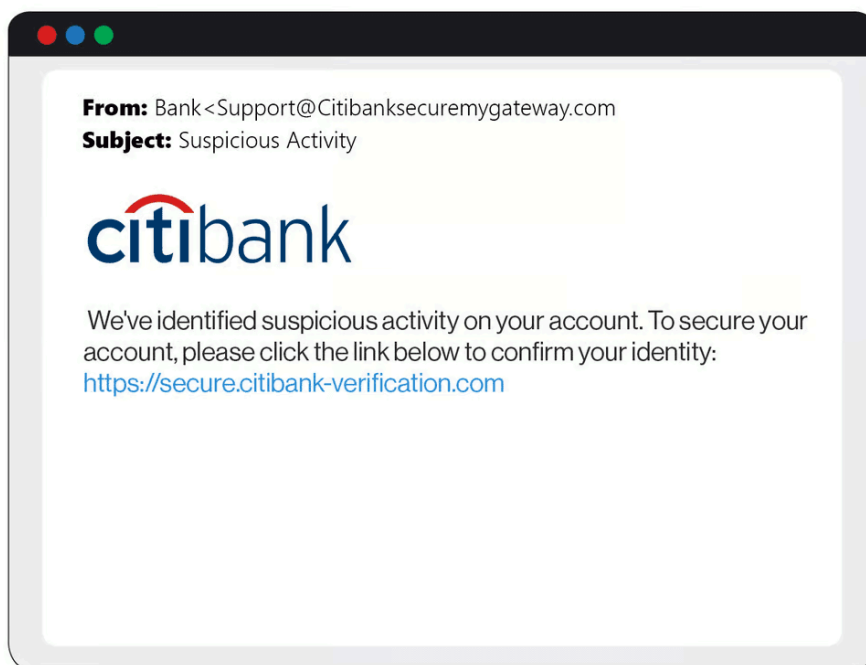
Netiquettes (Internet Etiquettes)

- Be polite and respectful in online communication.
- Don't use all caps (considered shouting).
- Don't spam or send chain messages.
- Respect privacy — don't share others' personal info without permission.
- Use proper language; avoid offensive content.

Security Considerations: Fraudulent & Unsolicited Emails (Phishing)

Phishing → Fraudulent attempt to steal sensitive info by pretending to be trustworthy (e.g., fake bank email asking for login).

Here are real-world examples of **phishing emails** (notice suspicious sender, urgent language, fake links):



keepnetlabs.com

● Wells Fargo

1:24 PM



Your Wells Fargo contact information has been updated

To: user@example.com

Danger (External, alerts@notify.wellsfargo.com)

Potential Sender Forgery, Brand Impersonation, and more... [Details](#)

[Report This Email](#) [FAQ](#) [Protection by Inky](#)



wellsfargo.com

We have updated your contact information

- Mailing Address

For details about what changed and to view the updates, or make additional updates [Download view or update contact information file](#).

If you have a questions, please call us at 1-800-869-3562 (for personal banking) or 1-800-225-5962 (for small business banking). We are available 24 hours a day, 7 days a week.

wellsfargo.com | [Fraud Information Center](#)

identityguard.com

How to recognize phishing:

- Unexpected/urgent requests for personal info.
- Suspicious sender email (e.g., bank@fake-domain.com).
- Hover over links — check real URL.
- Poor grammar or spelling mistakes.
- Never click links in suspicious emails — go directly to official site.

Basics of Software Licensing

- **Proprietary Software** → Paid, owned by company (e.g., Microsoft Windows, Adobe Photoshop). License restricts copying/use.
- **Freeware** → Free to use but may have restrictions.

- **Open Source** → Free, source code available (e.g., Linux, LibreOffice). You can modify/share.
- **Piracy** → Illegal copying/use of licensed software — can lead to fines/legal action.

Always use genuine software and check license terms.

Overview and Understanding of IT Act 2000 (India)

The **Information Technology Act, 2000** (amended in 2008) is India's primary law for cybercrimes and electronic commerce.

Key points:

- Legal recognition of electronic records and digital signatures.
- Defines cybercrimes like hacking, identity theft, phishing, spreading viruses.
- Section 66 → Hacking/punishment for unauthorized access.
- Section 66A (struck down in 2015) — previously covered offensive messages.
- Section 67 → Publishing obscene material electronically.
- Provides for **cyber appellate tribunal** and penalties/fines/imprisonment.
- Protects against data theft, cyber terrorism, and regulates intermediaries (e.g., social media platforms).

In simple terms: It makes online fraud, hacking, and misuse of digital data punishable in India.

Stay safe online — update software, use strong passwords, think before clicking, and report suspicious activity! If you have questions about any part (e.g., how to enable 2FA), feel free to ask, Sam. 😊